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United States Patent	12406826
Kind Code	B2
Date of Patent	September 02, 2025
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Charged particle beam device and sample observation method

Abstract

A charged particle optical system scans a sample with a pulsed charged particle beam and detects secondary charged particles; and a scan image is formed. Control is carried out so that a deflection signal for deflecting the charged particle beam in a first direction, a first timing for pulsed irradiation, a second timing for pulsed irradiation, and a third timing for detection of the secondary charged particles are synchronized. When the deflection amount of the charged particle beam in the time period of the first timing corresponds to the coordinates of n pixels in the scan image, the same line is scanned m times ($m < n$) while shifting the first timing with respect to the deflection signal so that a location irradiated with the charged particle beam by each scanning has different pixel coordinates. The pixel values at pixel coordinates where a signal is defective are restored.

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Appl. No.:	18/015605
Filed (or PCT Filed):	September 18, 2020
PCT No.:	PCT/JP2020/035486
PCT Pub. No.:	WO2022/059171
PCT Pub. Date:	March 24, 2022

Prior Publication Data

Document Identifier	Publication Date
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Publication Classification**Int. Cl.:** H01J37/28 (20060101); H01J37/20 (20060101); H01J37/244 (20060101)**U.S. Cl.:****CPC** H01J37/28 (20130101); H01J37/20 (20130101); H01J37/244 (20130101);**Field of Classification Search****CPC:** H01J (37/28); H01J (37/20); H01J (37/244); H01J (2237/2803); H01J (37/228); H01J (37/265); G01N (23/2251); G06N (3/09); G06N (20/00); H01L (22/00)

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Background/Summary

TECHNICAL FIELD

(1) The present disclosure relates to a pump-probe charged particle beam device that irradiates a sample with pump light or pump charged particles and then detects secondary charged particles generated from the sample based on irradiation of probe charged particles in a predetermined timing, and a sample observation method using the same.

BACKGROUND ART

(2) A charged particle beam device such as an electron microscope or an ion microscope is used for observation of various samples having a fine structure. For example, for a purpose of process management in a manufacturing process of a semiconductor device, a scanning electron microscope that is one of charged particle beam devices is applied to measurement such as dimension measurement or defect inspection of a semiconductor device pattern formed on a semiconductor wafer that is a sample.

(3) In recent years, three-dimensionalization of semiconductor devices and diversification of materials to be used have been developed, and there have been increasing needs for inspection and measurement of electrical characteristics and material characteristics. In response to such new needs, each of PTL 1 and PTL 2 discloses that a pump-probe electron microscope is effective, which acquires an image by irradiating a sample with pulsed light and then irradiating the sample with a pulsed electron beam.

(4) Specifically, PTL 1 discloses a method for observing a stacking fault in a sample by irradiation with a pulse laser having energy corresponding to a defect level in a semiconductor. PTL 2 discloses a method for acquiring an image with a high contrast by controlling a charged state, by adjusting, for a resist material used in an exposure process of manufacturing a semiconductor device, a time period of a pulse laser with which the resist material is irradiated.

CITATION LIST

Patent Literature

(5) PTL 1: WO2019/102603 PTL 2: WO2020/053967

SUMMARY OF INVENTION

Technical Problem

(6) In the scanning electron microscope having a pump-probe function disclosed in PTL 1 or PTL 2, detection signals cannot be acquired for all pixel positions by scanning with the electron beam once. This problem will be described with reference to FIGS. 1A and 1B. The scanning electron microscope forms a scan image by scanning a sample with primary electrons and detecting secondary electrons generated by scanning with the primary electrons.

(7) FIG. 1A is a time chart of a deflection signal **10** for scanning with primary electrons, a primary electron irradiation timing **11**, a detection sampling timing **12**, and pixel coordinates **13** in a normal scanning electron microscope (that does not use the pump-probe function). Since the primary electrons are continuously applied and detection sampling is performed at all the pixel coordinates, signals of secondary electrons can be detected at all pixel coordinates by scanning with the primary electrons once.

(8) FIG. 1B is a time chart of a deflection signal **20**, a pump light irradiation timing **21**, a probe electron irradiation timing **22**, a detection sampling timing **23**, and pixel coordinates **24** in the pump-probe scanning electron microscope. In the pump-probe scanning electron microscope, pump light for changing a state such as a charged state, an electron state, and a temperature of a sample is applied in a time period $T_{\text{sub.Pump}}$. Pump, probe electrons for forming a scan image are applied with a delay time $T_{\text{sub.Delay}}$ with respect to the pump light, and detection sampling is performed according to an irradiation timing of the probe electrons. Strictly speaking, since there is also a

delay time until the secondary electrons generated in the sample reach a detector, a delay time is formed between the irradiation timing of the probe electrons and the detection sampling timing, but the delay time is omitted here.

(9) Therefore, in the pump-probe scanning electron microscope, signals of the secondary electrons cannot be detected at all pixel coordinates by scanning the primary electrons only once, and in order to detect the signals of the secondary electrons at all pixel coordinates, it is necessary to scan with the primary electrons a plurality of times (at least eight times in this example) by shifting the pixel coordinates of the pump light and the probe electrons.

(10) As described above, the pump-probe scanning electron microscope is characterized in that an imaging time is longer than that of the normal scanning electron microscope. Therefore, when the pump-probe scanning electron microscope is used for dimension measurement or defect inspection of a semiconductor device pattern, a problem of a decrease in throughput is caused.

Solution to Problem

(11) According to an embodiment of the invention, a charged particle beam device includes: a charged particle optical system including a deflector, the charged particle optical system being configured to scan a sample with a pulsed charged particle beam by the deflector; a light source configured to irradiate the sample with pulsed light; a detector configured to detect secondary charged particles generated by irradiation on the sample with the charged particle beam; an image formation unit configured to form a scan image based on a signal from the detector; and a control unit configured to control the light source, the charged particle optical system, and the detector such that a deflection signal for deflecting the charged particle beam in a first direction, a first timing for irradiating the sample with the pulsed light, a second timing for irradiating the sample with the charged particle beam, and a third timing for detecting the secondary charged particles by the detector are synchronized.

(12) When a deflection amount of the charged particle beam in the first direction in a time period of the first timing corresponds to coordinates of n pixels in the scan image, the control unit scans a same line m times ($m < n$) while shifting the first timing with respect to the deflection signal such that a location irradiated with the charged particle beam by each scanning in the first direction has different pixel coordinates.

(13) The image formation unit restores a pixel value at pixel coordinates at which a signal is defective based on an integration scan image in which m scan images obtained by m times of scanning are integrated.

Advantageous Effects of Invention

(14) An imaging time of the charged particle beam device having a pump-probe function can be shortened.

(15) Other problems and novel features will be apparent from description of the present specification and the accompanying drawings.

Description

BRIEF DESCRIPTION OF DRAWINGS

- (1) FIG. 1A is a time chart of scan image formation in a normal scanning electron microscope.
- (2) FIG. 1B is a time chart of scan image formation in a pump-probe scanning electron microscope.
- (3) FIG. 2 is a configuration diagram of a scanning electron microscope according to a first embodiment.
- (4) FIG. 3 is a diagram showing scan image formation by integration.
- (5) FIG. 4 is an image restoration flow in a pump-probe scanning electron microscope.
- (6) FIG. 5 is an example of a condition setting screen.
- (7) FIG. 6 is a machine learning flow for image restoration.

(8) FIG. 7 is an example of an optimization condition setting screen.

(9) FIG. 8 is a time chart of the scan image formation for preventing occurrence of an artifact.

(10) FIG. 9 is a configuration diagram of a scanning electron microscope according to a second embodiment.

(11) FIG. 10 is a time chart of scan image formation in a pump-probe scanning electron microscope according to the second embodiment.

DESCRIPTION OF EMBODIMENTS

(12) Hereinafter, embodiments of the invention will be described. Although the drawings shown in the embodiments show specific embodiments according to the principle of the invention, these are for the purpose of understanding the invention and are not used to limit the invention. In the following embodiments, a scanning electron microscope using electrons as charged particles will be described as an example, but the same effect can be obtained even when various ions are used as charged particles.

First Embodiment

(13) FIG. 2 shows a configuration of a scanning electron microscope according to a first embodiment.

(14) The scanning electron microscope includes an electron beam optical system that irradiates a sample with an electron beam, a detection system that detects secondary electrons emitted from the sample due to irradiation with the electron beam, a stage mechanism system that is disposed in a vacuum chamber, a control system that controls components of the scanning electron microscope and processes various kinds of information, and an image processing system that executes processing such as image restoration on an obtained scan image.

(15) Specifically, primary electrons **102** generated by an electron source **101** are deflected by a deflector **104** and focused by an objective lens **103**, and then a sample **105** mounted on a movable stage **106** is irradiated with the primary electrons **102**. An operation of the objective lens **103** is controlled by an objective lens control unit **113**, an operation of the deflector **104** is controlled by a deflector control unit **114**, and an operation of the movable stage **106** is controlled by a stage control unit **107**. A negative voltage may be applied to the sample **105** via the movable stage **106**.

(16) In order to add a pump-probe function to the scanning electron microscope, the primary electrons **102** need to be pulsed. This is implemented by a blanking electrode **108** and an aperture **109**. When a voltage is applied to the blanking electrode **108**, the primary electrons **102** are deflected and collide with the aperture **109**. On the other hand, when no voltage is applied to the blanking electrode **108**, the primary electrons **102** pass through the aperture **109** and are applied to the sample **105**. Therefore, the primary electrons **102** can be pulsed by controlling the voltage applied to the blanking electrode **108**. An operation of the blanking electrode **108** is controlled by a blanking control unit **110**. In this example, the primary electrons **102** are pulsed by the blanking electrode **108**, but a method for pulsing the primary electrons **102** is not limited thereto. For example, the electron source **101** may be a photocathode and the electron source **101** may be irradiated with a pulsed laser, or a voltage (not shown) for extracting the primary electrons **102** from the electron source **101** may be pulsed.

(17) Pulsed secondary electrons **111** generated by irradiation on the sample **105** with the primary electrons **102** pulsed by the blanking electrode **108** and the aperture **109** are detected by a detector **112**. Although the detector **112** is disposed closer to the electron source **101** than is the deflector **104** in the configuration shown in FIG. 2, the detector **112** may be disposed between the deflector **104** and the objective lens **103** or between the objective lens **103** and the sample **105** as long as the detector **112** can detect the secondary electrons **111**. Examples of a configuration of the detector **112** include an E-T detector and a semiconductor detector each including a scintillator, a light guide, and a photomultiplier tube, but any detector may be used as long as the detector **112** has a configuration capable of detecting electrons. The detector **112** may be mounted at a plurality of positions. A timing at which a signal for forming a scan image is acquired from the detector **112** is

controlled by a detection sampling control unit **115**.

(18) An image formation unit **116** assigns the signal acquired by the detection sampling control unit **115** to pixel coordinates of the primary electrons **102** determined by the deflector control unit **114** to form a scan image. The generated scan image is displayed on an image display unit **117** and recorded in a recording unit **118**.

(19) The electron source **101**, the blanking electrode **108**, the aperture **109**, the detector **112**, the deflector **104**, the objective lens **103**, the sample **105**, and the movable stage **106** are accommodated in a housing **119**, and the inside of the housing **119** is maintained in a vacuum state by a vacuum pump (not shown).

(20) In order to change a state of the sample **105**, a pulsed laser **120** is installed outside the housing **119**, and pulsed light **121** emitted from the pulsed laser **120** passes through a port **122** attached to the housing **119** and is applied to the sample **105**. Here, a wavelength of the pulsed light **121** is typically in a range of ultraviolet light to visible light, but a microwave, a terahertz wave, an X-ray, or a y-ray may be used as long as the light is pulsed. An irradiation timing of the pulsed light **121** generated by the pulsed laser **120** is controlled by a pulsed laser control unit **123**.

(21) Operations of the stage control unit **107**, the blanking control unit **110**, the objective lens control unit **113**, the deflector control unit **114**, the detection sampling control unit **115**, the image formation unit **116**, the image display unit **117**, the recording unit **118**, the pulsed laser control unit **123**, and an image processing unit **124** to be described later are controlled by a workstation (control unit) **125**.

(22) A method for executing image restoration in the pump-probe scanning electron microscope shown in FIG. 2 will be described. As shown in FIG. 1B, in the pump-probe scanning electron microscope, signals of all pixel coordinates cannot be detected only by scanning once. For example, when the pulsed light **121** and the primary electrons **102** are applied every four pixels, it is necessary to scan four times while shifting an irradiation timing of the pulsed light **121** and the primary electrons **102** with respect to a deflection signal by one pixel. At this time, a relationship between a scan image (schematic diagram) obtained by each scanning and a finally obtained scan image (schematic diagram) is shown in FIG. 3. Black vertical lines appearing in scan images **31** to **34** indicate that no signal is recorded, and it can be confirmed that each of the scan images **31** to **34** has signals only in discrete pixels. In the related-art pump-probe scanning electron microscope, the first to fourth scan images **31** to **34** obtained by scanning four times are integrated to obtain a scan image (integration scan image) **35** in which signals of all pixel coordinates are recorded. When the pump-probe scanning electron microscope scans a plurality of times by shifting the irradiation timing of the pulsed light **121** and the primary electrons **102** with respect to the deflection signal, each line or each frame may be repeatedly scanned. The same applies to the following embodiments and modifications.

(23) On the other hand, in the present embodiment, in order to reduce the number of times of scanning and improve a throughput of the pump-probe scanning electron microscope, image restoration is executed by predicting signals of pixels in which no signal is recorded based on an image in which signals of only in a part of pixels are recorded. FIG. 4 shows a specific flow. After a sample is inserted, a time period $T_{\text{sub.Pump}}$ of the pulsed light **121** is set (**S402**), and a delay time $T_{\text{sub.Delay}}$ of the primary electrons **102** with respect to the pulsed light **121** is set (**S403**). Next, a thinning ratio P of sampling pixels is set (**S404**). For example, when the time period $T_{\text{sub.Pump}}$ is set to correspond to four pixels, electron beam scanning of four times are required in order to obtain signals in all pixels (thinning ratio $P=0$). When the thinning ratio P is set to 25%, in an example shown in FIG. 3, in order to obtain any three scan images among the scan images **31** to **34**, the electron beam scanning is performed three times.

(24) Various observation conditions including conditions set in steps **S402** to **S404** can be set via a GUI displayed on the image display unit **117**. FIG. 5 shows an example of a condition setting screen. A condition setting screen **40** includes a scan image display section **41** and an observation

condition setting section **42**. The observation condition setting section **42** includes an observation condition setting section **43** set in steps **S402** to **S404**. A normal scan image (without using the pump-probe function) acquired by the pump-probe scanning electron microscope or a pump-probe scan image can be displayed in the scan image display section **41**. Thereby, search in an observation field of view and adjustment on the observation conditions including an image restoration condition can be performed while confirming the image in the scan image display section **41**.

(25) Next, a scan image is acquired under the set observation conditions (**S405**). Then, signals of pixels in which no signal is recorded are restored based on an integration scan image in which scan images acquired by the image processing unit **124** are integrated (**S406**), and the restored image (restored scan image) is displayed on the image display unit **117** (**S407**).

(26) Here, image restoration executed in **S406** can be executed using a dictionary stored in the recording unit **118**. It is known that non-defective image data can be obtained by restoration from image data in which pixel data is partially defective by sparse modeling. By storing, in the recording unit **118**, base images (a set of base images is referred to as the dictionary) used for sparse modeling trained using a pump-probe scan image acquired in advance, it is possible to estimate signals of all pixels of the pump-probe scan image based on the pump-probe scan image in which some pixels are thinned out. Alternatively, the image restoration may be executed using a processing algorithm of machine learning stored in the recording unit **118**. A specific learning method in this case will be described later. By using either method, it is possible to reduce an image acquisition time of the scan image according to the thinning ratio P of sampling pixels, as compared with the related-art pump-probe scanning electron microscope.

(27) FIG. **6** shows a flow of machine learning for image restoration. The machine learning according to the present embodiment is supervised learning, and requires a scan image in which signals of all pixels are recorded as training data. Therefore, after the time period $T_{\text{sub.Pump}}$ (**S602**) of the pulsed light **121** and the delay time $T_{\text{sub.Delay}}$ (**S603**) of the primary electrons **102** are set, the sample **105** is moved to an imaging position by the movable stage **106** (**S604**), and a scan image is acquired (**S605**). The scan image acquired in step **S605** is an integration scan image in which signals are recorded in all pixels, which is obtained by scanning a plurality of times at the thinning ratio $P=0\%$ and integrating scan images. Movement of the imaging position and acquisition of the scan image are repeated a predetermined number of times to acquire a sufficient number of scan images for learning (**S606**). The acquired scan image is recorded in the recording unit **118**. Thereafter, the thinning ratio P of sampling pixels is set (**S607**), and a thinned scan image is generated from the scan image acquired in step **S605** according to the set thinning ratio P (**S608**). Then, machine learning is performed such that the scan image acquired in step **S605** can be reproduced from the thinned scan image generated in step **S608** (**S609**). Here, a deep neural network, convolutional neural networks, generative adversarial networks, and the like can be used as an example of a learning algorithm. In addition, any algorithm can be applied as long as the algorithm can estimate, based on a thinned scan image, a scan image in which signals of all pixels are recorded. A trained model obtained by the machine learning is stored in the recording unit **118** (**S610**), and the flow ends. Signals of a defective signal portion are restored using the trained model stored in the recording unit **118** (**S406**). Instead of creating a scan image with the predetermined thinning ratio P in a pseudo manner based on an integration scan image with the thinning ratio $P=0\%$, an integration scan image acquired while actually changing the thinning ratio P may be used for learning.

(28) Here, in the pump-probe scanning electron microscope, a contrast of the obtained scan image is changed by changing the time period $T_{\text{sub.Pump}}$ of the pulsed light **121** and the delay time $T_{\text{sub.Delay}}$ of the primary electrons **102**. When the thinning ratio P of sampling pixels is increased, an image acquisition time is shortened, but a probability that an artifact, which is an image not actually existing in the restored image, is mixed increases.

(29) Therefore, it is desirable that a user can search for an observation condition that provides an optimum contrast and an optimum image acquisition time. Therefore, in the pump-probe scanning electron microscope according to the present embodiment, a scan image obtained by executing image restoration under a plurality of conditions for the time period $T_{\text{sub.Pump}}$ of the pulsed light **121**, the delay time $T_{\text{sub.Delay}}$ of the primary electrons **102**, and the thinning ratio P of sampling pixels is displayed on the image display unit **117**. FIG. 7 shows an example of an optimization condition setting screen. The optimization condition setting screen **50** may be called, for example, by providing an optimization button on the condition setting screen **40** shown in FIG. 5 and pressing the optimization button. In this case, it is desirable to set a plurality of conditions based on the time period $T_{\text{sub.Pump}}$ of the pulsed light **121** and the delay time $T_{\text{sub.Delay}}$ of the primary electrons **102** set in the observation condition setting section **43** of the condition setting screen **40**. The user may directly set the plurality of conditions.

(30) In general, a contrast of the scan image tends to change when the time period $T_{\text{sub.Pump}}$ of the pulsed light **121** and the delay time $T_{\text{sub.Delay}}$ of the primary electrons **102** are changed, and artifacts tend to increase when the thinning ratio P is increased. Therefore, in the example shown in FIG. 7, three types of combinations of the time period $T_{\text{sub.Pump}}$ of the pulsed light **121** and the delay time $T_{\text{sub.Delay}}$ of the primary electrons **102** and three types of thinning ratios P are set, and nine pump-probe images are acquired in total by the combinations and displayed on the optimization condition setting screen **50**. Each column has the same combination of the time period $T_{\text{sub.Pump}}$ of the pulsed light **121** and the delay time $T_{\text{sub.Delay}}$ of the primary electrons **102**, and the thinning ratios P are set to 0%, 25%, and 50% from the top. As a schematic diagram, in the pump-probe image, a circle represents a pattern, a difference in color between the circle and a background represents a contrast, and a deviation in a shape of the circle represents an artifact.

(31) The user selects a desired scan image from the scan images displayed on the optimization condition setting screen **50** in consideration of a relationship between the contrast and the artifact. The time period $T_{\text{sub.Pump}}$ of the pulsed light **121**, the delay time $T_{\text{sub.Delay}}$ of the primary electrons **102**, and the thinning ratio P of sampling pixels of the selected scanned image are set as conditions for acquiring the scan image.

(32) In this example, the number of conditions for acquiring the pump-probe image is nine, but the number is not limited thereto. First, the thinning ratio P may be fixed to 0%, and the conditions of the time period $T_{\text{sub.Pump}}$ of the pulsed light **121** and the delay time $T_{\text{sub.Delay}}$ of the primary electrons **102** may be changed to search for conditions for obtaining a scan image with a good contrast, and then, the two conditions may be fixed and only the thinning ratio P may be changed to finally set the three conditions. This makes it possible to efficiently search for conditions for obtaining a good pump-probe image.

(33) Further, a method for reducing artifacts during image restoration in the pump-probe scanning electron microscope will be described. In general, in a process of restoring an original image based on a sparse sampling image, an artifact is less likely to appear when sampling is random. However, in the pump-probe scanning electron microscope, the scan image needs to be acquired with a plurality of time periods $T_{\text{sub.Pump}}$ of the pulsed light **121** and the delay time $T_{\text{sub.Delay}}$ of the primary electrons **102** kept constant. Therefore, sampling intervals are equal, and there is a tendency that an artifact is likely to occur in the restored image.

(34) FIG. 8 shows a time chart of a deflection signal **60** for scanning primary electrons, a pump light irradiation timing **61**, a probe electron irradiation timing **62**, a detection sampling timing **63**, and pixel coordinates **64** for preventing occurrence of an artifact. Although FIG. 8 shows an example in which the primary electrons **102** are emitted for scanning in a horizontal direction, the same effect can be attained even when the primary electrons **102** are emitted for scanning in a vertical direction. During scanning of a first line, the time period $T_{\text{sub.Pump}}$ of the pulsed light **121** and the delay time $T_{\text{sub.Delay}}$ of the primary electrons **102** are fixed, and detection sampling is also performed at equal intervals. During scanning of a second line, the time period $T_{\text{sub.Pump}}$

of the pulsed light **121** and the delay time T.sub.Delay of the primary electrons **102** are the same as those of the first line, but an irradiation start timing of the pulsed light **121** is set to a timing different from that of the first line, and the same applies to the third and subsequent lines. As described above, the time period T.sub.Pump of the pulsed light **121**, the delay time T.sub.Delay of the primary electrons **102**, and a shift amount change pattern of an irradiation timing of the pulsed light **121** with respect to the deflection signal **60** for each m times of scanning are the same in a plurality of lines, while an initial shift amount of the irradiation timing of the pulsed light **121** with respect to the deflection signal **60** is different for each of the plurality of lines, whereby it is possible to perform sampling with increased randomness in a direction orthogonal to a scanning direction of the primary electrons **102** while fixing the time period T.sub.Pump of the pulsed light **121** and the delay time T.sub.Delay of the primary electrons **102**.

(35) An irradiation start timing of the pulsed light **121** in each line may be determined by the workstation (control unit) **125** generating a random number when scanning of each line is started, or the irradiation start timing may be controlled by reading an irradiation start timing recorded in the recording unit **118** in advance. Since occurrence of an artifact is affected by a shape of a pattern, a scan image or a design pattern of an observation target may be input, a correspondence table of an irradiation start timing corresponding to a predetermined pattern may be stored in advance in the recording unit **118**, and an irradiation start timing may be determined by comparing the correspondence table with the input scan image or design pattern. Alternatively, a trained model in which a pattern and a preferable irradiation start timing are trained may be created, and an irradiation start timing may be obtained using the trained model.

Second Embodiment

(36) In the first embodiment, an example in which a pulsed laser is used as a pump source has been described. However, it is effective even when a charged particle source of an electron beam or the like is used as a pump source, and a configuration in which a charged particle source is used as a pump source will be described as a second embodiment.

(37) FIG. **9** shows a configuration of a scanning electron microscope according to the second embodiment. The same components as those according to the first embodiment are denoted by the same reference numerals, and repeated description thereof will be omitted. In the configuration shown in FIG. **9**, the pulsed laser **120**, the port **122**, and the pulsed laser control unit **123** for irradiation of pump light are removed from the configuration shown in FIG. **2**, a pulsed electron source **901** is attached to the housing **119**, and the sample **105** is irradiated with pulsed electrons **902** generated by the pulsed electron source **901**. The pulsed electron source **901** is controlled by a pulsed electron source control unit **903**. A charged particle beam source that irradiates another pulsed charged particle beam such as ions may be mounted instead of the pulsed electron source **901**.

(38) A detection sampling method in a case of using an electron beam as the pump source will be described with reference to FIG. **10**. FIG. **10** is a time chart of a deflection signal **70**, a pump electron irradiation timing **71**, a probe electron irradiation timing **72**, a detection sampling timing **73**, and pixel coordinates **74** in the pump-probe scanning electron microscope according to the second embodiment. When electrons are used as the pump source, the secondary electrons **111** are generated from the sample **105** by both irradiation of pump electrons and irradiation of probe electrons, and by matching a detection sampling timing with a probe electron irradiation timing, the secondary electrons **111** generated by the probe electrons are detected and a scan image is formed.

(39) Irradiation energy of the pump electrons and the probe electrons to the sample **105** may be the same or different. As a method for changing the irradiation energy of the pump electrons, an acceleration voltage of the pulsed electron source **901** is set to a value different from that of the electron source **101**, or a voltage (retarding voltage) applied to the sample **105** is changed at the time of irradiation of the pump electrons and the probe electrons. It is desirable that a spot diameter of the pump electrons is set to a sufficient size for an observation field of view such that a probe

current amount does not greatly vary for each pixel coordinate.

(40) The electron source **101** may also be used as a pump electron source. A configuration of the scanning electron microscope in this case is obtained by removing the pulsed laser **120**, the port **122**, and the pulsed laser control unit **123** from the configuration shown in FIG. **2**. As in a case of the probe electrons, the pump electrons are also pulsed by pulsing the primary electrons **102** by the blanking electrode **108** and the aperture **109**. The pump electrons may be defocused from the probe electrons. However, when a distance between an irradiation location of the pump electrons and an irradiation location of the probe electrons is equivalent to several pixels, an influence of the pump electrons can be generally observed at the irradiation location of the probe electrons without changing irradiation conditions of the primary electrons **102**, although depending on a pattern to be observed.

(41) The invention has been described above with reference to the embodiments and modifications. The embodiments and modifications described above may be modified in various ways without departing from the scope of the invention, and may be used in combination.

REFERENCE SIGNS LIST

(42) **10, 20, 60, 70**: deflection signal **11**: primary electron irradiation timing **12, 23, 63, 73**: detection sampling timing **13, 24, 64, 74**: pixel coordinates **21, 61**: pump light irradiation timing **22, 62, 72**: probe electron irradiation timing **31 to 35**: scan image **40**: condition setting screen **41**: scan image display section **42, 43**: observation condition setting section **50**: optimization condition setting screen **71**: pump electron irradiation timing **101**: electron source **102**: primary electrons **103**: objective lens **104**: deflector **105**: sample **106**: movable stage **107**: stage control unit **108**: blanking electrode **109**: aperture **110**: blanking control unit **111**: secondary electrons **112**: detector **113**: objective lens control unit **114**: deflector control unit **115**: detection sampling control unit **116**: image formation unit **117**: image display unit **118**: recording unit **119**: housing **120**: pulsed laser **121**: pulsed light **122**: port **123**: pulsed laser control unit **124**: image processing unit **125**: workstation (control unit) **901**: pulsed electron source **902**: pulsed electrons **903**: pulsed electron source control unit

Claims

1. A charged particle beam device comprising: a charged particle optical system including a deflector, the charged particle optical system being configured to scan a sample with a pulsed charged particle beam by the deflector; a light source configured to irradiate the sample with pulsed light; a detector configured to detect secondary charged particles generated by irradiation on the sample with the charged particle beam; an image formation unit configured to form a scan image based on a signal from the detector; and a control unit configured to control the light source, the charged particle optical system, and the detector such that a deflection signal for deflecting the charged particle beam in a first direction, a first timing for irradiating the sample with the pulsed light, a second timing for irradiating the sample with the charged particle beam, and a third timing for detecting the secondary charged particles by the detector are synchronized, wherein when a deflection amount of the charged particle beam in the first direction in a time period of the first timing corresponds to coordinates of n pixels in the scan image, the control unit scans a same line m times ($m < n$) while shifting the first timing with respect to the deflection signal such that a location irradiated with the charged particle beam by each scanning in the first direction has different pixel coordinates, and the image formation unit restores a pixel value at pixel coordinates at which a signal is defective based on an integration scan image in which m scan images obtained by m times of scanning are integrated.

2. The charged particle beam device according to claim 1, further comprising: an image display unit, wherein the control unit displays, on the image display unit, a restored scan image in which the pixel value at the pixel coordinates at which the signal is defective is restored.

3. The charged particle beam device according to claim 2, wherein the control unit receives setting of the time period of the first timing, a delay time from the first timing to the second timing, and a thinning ratio, and sets a value of m according to the thinning ratio.
4. The charged particle beam device according to claim 3, wherein in setting the time period of the first timing and the delay time from the first timing to the second timing, the control unit displays, on the image display unit, a plurality of integration scan images, in each of which n scan images obtained by scanning at a thinning ratio of 0% are integrated, under a plurality of conditions in which the time period of the first timing and the delay time from the first timing to the second timing are different from each other.
5. The charged particle beam device according to claim 4, wherein the control unit displays, on the image display unit, a plurality of restored scan images restored based on integration scan images, in each of which m scan images obtained by scanning at a different thinning ratio are integrated, under conditions including the set time period of the first timing and the set delay time from the first timing to the second timing.
6. The charged particle beam device according to claim 1, wherein the control unit scans a plurality of lines by moving a scanning start position in the first direction to a second direction orthogonal to the first direction, and the control unit varies an initial shift amount of the first timing with respect to the deflection signal for each of the plurality of lines.
7. The charged particle beam device according to claim 6, wherein the initial shift amount of the first timing with respect to the deflection signal is set to be random or a value set in advance for each of the plurality of lines.
8. The charged particle beam device according to claim 1, further comprising: a recording unit configured to store a dictionary in which a base image is recorded, wherein the image formation unit restores, using the dictionary, the pixel value at the pixel coordinates at which the signal is defective, by sparse modeling, based on the integration scan image in which the m scan images are integrated.
9. The charged particle beam device according to claim 1, further comprising: a recording unit configured to store a trained model, wherein the image formation unit restores, using the trained model, the pixel value at the pixel coordinates at which the signal is defective based on the integration scan image in which m scan images are integrated.
10. The charged particle beam device according to claim 1, further comprising: instead of the light source, a charged particle source configured to irradiate the sample with a charged particle beam pulsed in the first timing.
11. The charged particle beam device according to claim 10, wherein the charged particle source is shared with a charged particle source of the charged particle optical system.
12. A sample observation method using a charged particle beam device, the charged particle beam device including: a charged particle optical system including a deflector, the charged particle optical system being configured to scan a sample with a pulsed charged particle beam by the deflector; a light source configured to irradiate the sample with pulsed light; a detector configured to detect secondary charged particles generated by irradiation on the sample with the charged particle beam; an image formation unit configured to form a scan image based on a signal from the detector; a control unit configured to control the light source, the charged particle optical system, and the detector such that a deflection signal for deflecting the charged particle beam in a first direction, a first timing for irradiating the sample with the pulsed light, a second timing for irradiating the sample with the charged particle beam, and a third timing for detecting the secondary charged particles by the detector are synchronized; and an image display unit, the sample observation method comprising: by the control unit, when a deflection amount of the charged particle beam in the first direction in a time period of the first timing corresponds to coordinates of n pixels in the scan image, scanning a same line m times ($m < n$) while shifting the first timing with respect to the deflection signal such that a location irradiated with the charged

particle beam by each scanning in the first direction has different pixel coordinates; by the image formation unit, restoring a pixel value at pixel coordinates at which a signal is defective based on an integration scan image in which m scan images obtained by the m times of scanning are integrated; and by the control unit, displaying, on the image display unit, a restored scan image in which the pixel value at the pixel coordinates at which the signal is defective is restored.

13. The sample observation method according to claim 12, wherein the control unit receives setting of the time period of the first timing, a delay time from the first timing to the second timing, and a thinning ratio from a condition setting screen displayed on the image display unit, and sets a value of m according to the thinning ratio.

14. The sample observation method according to claim 13, wherein in setting the time period of the first timing and the delay time from the first timing to the second timing, the control unit displays, on the image display unit, a plurality of integration scan images, in each of which n scan images obtained by scanning at a thinning ratio of 0% are integrated, under a plurality of conditions in which the time period of the first timing and the delay time from the first timing to the second timing are different from each other.

15. The sample observation method according to claim 12, wherein the control unit scans a plurality of lines by moving a scanning start position in the first direction to a second direction orthogonal to the first direction, and the control unit sets, for each of the plurality of lines, an initial shift amount of the first timing with respect to the deflection signal to be random for each of the plurality of lines, or a value set in advance to be different for each of the plurality of lines.
